



GENERAL SIR JOHN KOTALAWALA DEFENCE UNIVERSITY

Faculty of Technology

Department of Biosystems Technology

**BT (Hons) in Information and Communication Technology Degree
Programme.**

Professional Ethics in IT

PROFESSIONAL ETHICS IN IT

Responsibilities and Rights

Individual Assessment

3rd Year ICT Students

Duration:	3 Hours
Total Marks:	100
Student Name:	
Student ID:	

Instructions to Students

This assignment assesses your ability to analyze ethical dilemmas, apply ethical frameworks, and understand professional responsibilities in the IT sector. You will work with real-world case studies, Sri Lankan legal contexts, and conduct primary research through interviews.

Academic Integrity Notice

This assignment requires original thinking, research, and analysis. AI-generated responses without personal research, critical analysis, and proper evidence will receive a failing grade. Assignment will be checked through Turnitin for AI percentage and plagiarism.

You MUST include:

- Screenshots of your evidence process (Google searches, website visits)
- Completed interview forms with signatures from interviewees
- Direct quotes from Sri Lankan Acts with proper citations (Act name, section number, year)
- Hand-drawn diagrams or flowcharts (photograph and include)

Submission Requirements

Submit a single PDF document containing:

1. Main report (3,500-4,500 words)
2. Appendices with all required evidence (interview forms, research screenshots)
3. Photographs of hand-drawn diagrams/flowcharts

PART A: ETHICAL FRAMEWORKS

Question 1: Respect for Authority in IT Organizations

Scenario:

A junior engineer notices a serious security problem in a live system. Thinking it is urgent, they fix it immediately without getting approval. The problem is solved, but the system later stops working properly and customers are affected.

Required Analysis:

- (a) Explain what went wrong in this scenario. Identify at least FOUR specific mistakes made by the junior engineer.
- (b) Describe the difference between 'respecting authority' and 'blind obedience'. Use this scenario to illustrate your explanation.
- (c) Research and document TWO real-world cases where IT professionals faced consequences for either: (i) Following orders that led to harm, OR (ii) Disobeying orders to prevent harm. Provide case names, dates, companies, and outcomes.
- (d) Create a flowchart showing the proper escalation process the junior engineer should have followed. Your flowchart must include: Initial discovery, Risk assessment, Decision points, Escalation paths, Documentation requirements. (Hand-draw and photograph, or use a digital tool and take screenshot)

Evidence Required: Research screenshots, flowchart photograph/screenshot

Question 2: Confidentiality and Data Protection in Sri Lanka

Real-World Case Study: Equifax Data Breach (2017)

Between May and July 2017, Equifax was breached. Private records of 147.9 million Americans, 15.2 million British citizens, and 19,000 Canadian citizens were compromised, making it one of the largest cybercrimes related to identity theft.

Required Analysis:

- (e) Research the Equifax breach and answer: What was the technical vulnerability? How long did Equifax know about it before the breach? When did they disclose it to the public? What were the consequences?
- (f) Analyze this breach using the Sri Lanka Personal Data Protection Act No. 9 of 2022. If Equifax had been a Sri Lankan company: What sections of the Act would apply? What penalties could they face? What notification requirements would they have? ~~(Quote specific sections with citations)~~
- (g) According to the presentation slide 'Ethical Questions After a Data Breach', four key questions were raised. Answer each question from the perspective of:

- (h) (i) An IT professional who discovered the breach,
- (i) (ii) The company CEO,
- (j) (iii) An affected customer. Explain any conflicts between these perspectives.

Question 3: Conflicts of Interest

Required Tasks:

- (k) Define 'conflict of interest' in your own words. Explain the difference between: Actual conflict, Potential conflict, Perceived conflict. Provide one original example of each in an IT context.
- (l) Create a 'Conflict of Interest Decision Matrix' for IT students. Include at least 6 scenarios such as: Working on a group project where your best friend is in the group, Being offered a part-time job at a company where your family member works, Receiving gifts from a software vendor for a university project, Using university resources for personal freelance work, Recommending a friend's startup in your final year project, Having romantic relationships within project teams.

For each scenario, classify as: Acceptable / Requires Disclosure / Should Be Avoided, and justify your reasoning.

- ~~(m)~~ Research and document ONE real case of conflict of interest in the global IT industry (e.g., board members, executives, engineers). Describe: What the conflict was, How it was discovered, What the consequences were, What lessons IT professionals should learn. ~~(Provide proper citations)~~

Question 4: Employee Rights in Sri Lanka

Required Research and Analysis:

- (n) Research the following Sri Lankan legislation relevant to IT workers: Shop and Office Employees Act (working hours, overtime), Industrial Disputes Act (dispute resolution), Termination of Employment of Workmen Act (notice periods, termination). For each Act, summarize: Key protections for employees, Penalties for violations, One specific section that IT workers should know.
- (o) Create a 'Know Your Rights' infographic (a diagram) for IT workers in Sri Lanka.

Include: Minimum wage/salary protections, Working hours limits, Leave entitlements, Termination notice requirements, Where to report violations.

(Hand-draw or use digital tool, include visual elements)

PART B: CASE STUDY ANALYSIS

Case Study: The Deployment Dilemma

Background:

You are Nimesh, a QE Lead at 'TechSolutions Lanka', a software company in Colombo developing a mobile banking application for a major Sri Lankan bank. Your team of 6 testers has been working on this critical project for 8 months.

The Situation:

Three days before the scheduled production deployment, during final regression testing, your team discovers a critical security vulnerability:

- The app's password reset function can be exploited to gain unauthorized access to user accounts*
- The vulnerability affects the fund transfer module, potentially exposing customer bank accounts*
- This bug has existed in the code for 4 months but was only discovered due to new test scenarios*
- Fixing it properly will take at least 2 weeks including testing*

The Pressure:

Your Project Manager, Kasun, calls an urgent meeting. Present are:

- Kasun (Project Manager) - under intense pressure from senior management*
- Hiruni (Development Lead) - says it's 'just a theoretical risk, unlikely to be exploited'*
- You (QE Lead) - responsible for quality and security sign-off*
- Dilshan (CEO) - joins virtually, emphasizes business implications*

Kasun presents the situation:

- 'The bank has already announced the launch to their 2 million customers'*
- 'We'll face liquidated damages of Rs. 15 million for delay'*
- 'The team has been working 60-hour weeks to meet this deadline'*
- 'No actual exploitation has occurred in 4 months'*
- 'We can deploy now and fix it in the next release in 3 weeks'*

Dilshan (CEO) adds:

- 'This contract is 40% of our annual revenue'*
- 'Delay could damage our reputation with the bank'*
- 'I need your professional assessment, Nimesh, but understand the stakes'*

Additional Complications:

- Your annual performance review is next week*

- *Two of your team members have already raised concerns about the bug in writing*
- *The bank's IT security team hasn't been informed about this finding yet*
- *You know that if money is stolen due to this vulnerability, customers could lose their savings*
- *You're aware of Sri Lanka's Personal Data Protection Act but unsure of exact penalties*

Your Decision:

Kasun looks at you and asks: 'Nimesh, as QE Lead, do you sign off on this deployment? Yes or no?'

Required Analysis

Section 1: Ethical Analysis

- (p) Identify ALL ethical issues present in this scenario. Categorize them under: Respect for Authority, Confidentiality, Conflict of Interest, Professional Responsibility, Employee Rights. (Minimum 8 issues identified)
- (q) Analyze the ethical positions of each stakeholder: Nimesh (you), Kasun (PM), Hiruni (Dev Lead), Dilshan (CEO), The bank, End customers, Your QE team members. For each stakeholder, identify: Their primary interest, Their ethical responsibility, Potential conflicts between interest and responsibility.

Section 2: Legal and Regulatory Analysis

- (r) Research and mention specific sections of the Sri Lanka Personal Data Protection Act No. 9 of 2022 that apply to this scenario. Address: Data breach notification requirements, Penalties for security failures, Controller vs. Processor responsibilities, Rights of data subjects.
- (s) Research if there are any other Sri Lankan laws or regulations that apply (e.g., banking regulations, computer crime laws). ~~Cite specific legislation.~~
- (t) Compare this scenario to the Equifax breach discussed in class. Identify 3 similarities and 3 differences. What lessons from Equifax apply here?

Section 4: Alternative Strategies

Develop THREE different response strategies:

Strategy A: Deploy Now (As management suggests)

- Specific actions you would take
- Risk mitigation measures
- Documentation requirements
- Ethical and legal implications

Strategy B: Refuse to Deploy

- How you would communicate this decision
- Escalation path you would follow
- How to protect yourself and your team
- Consequences you might face

Strategy C: Negotiate a Middle Ground

- What compromises you would propose
- Conditions for limited deployment
- Stakeholder communication plan
- Feasibility and risks

Section 5: Your Recommendation and Justification

Clearly state which strategy you would choose and provide detailed justification covering:

- Why this is the most ethical choice
- How it balances competing interests
- How it complies with legal requirements
- What professional values it upholds

Answers